

SMART INDIA HACKATHON 2026



DEPTHWIZARD

One satellite image in, a measurable 3D surface out.

PROBLEM STATEMENT ID

SIH26175

PROBLEM STATEMENT TITLE

Single-View Height Estimation and 3D
Terrain Visualisation

THEME

Space Technology

PS CATEGORY

Software

TEAM ID

<fill from portal>

TEAM NAME (ON PORTAL)

Dev Up

BUILT AND MEASURED, NOT PROPOSED

3.62 m

per-building RMSE
against airborne LiDAR

39 %

better than the 5.9 m
GlobalBuildingAtlas bar

509 ms

per tile on CPU alone,
no GPU in the loop



The palette is the model's own output scale: 0 m at the left, 155 m at the right.



THE PROBLEM

01

Archives with no height

Decades of Cartosat single-view imagery exist. Stereo and LiDAR cannot be flown into the past.

02

Answers with no confidence

Single-view height is ill-posed. One number implies a certainty the model does not have.

03

Checkpoints are not products

The literature ships weights and an eval script. An analyst needs a scene they can open and check.

OUR SOLUTION



One image to a metric DSM

Depth Anything V2 fine-tuned on DFC2019 LiDAR. GeoTIFF out, metadata or not.



Confidence, not just height

A per-pixel sigma beside every height. Where it says unsure, it is wrong — monotonically.



A navigable 3D scene

three.js flythrough, drag-to-compare against reference, and a live error map.



Validation inside the product

RMSE, MAE and correlation against LiDAR — per terrain class, not one flattering average.



Absolute heights, two ways

Copernicus GLO-30, or ground control points via a power-law fit.



Deployable, not demo-ware

No GPU: 509 ms per tile. The viewer is one 15 MB HTML file that opens from disk.

THE PRODUCT, RUNNING — CAPTURES OF THE LIVE VIEWER, NOT MOCK-UPS

SIKKIM, INDIA



967 m of relief across 2 km, image draped on the surface.

OMAHA, HEIGHT ABOVE GROUND



Flat roofs, tree crowns and kerbs resolved at 0.3 m.

OMAHA, MODEL CONFIDENCE



Per-pixel sigma: where the estimate is weak, it says so.

ARCHITECTURE — ONE IMAGE TO A CHECKED SURFACE

- 1 INPUT**
GeoTIFF, or a plain PNG / JPG with no metadata at all
- 2 GSD NORMALISE**
auto-zoom reads ground sample distance, resamples to 0.3 m
- 3 DA-V2 BACKBONE**
ViT encoder and DPT neck, fine-tuned on DFC2019 LiDAR
- 4 DUAL HEAD**
height mu and log-variance sigma, emitted per pixel
- 5 TILED INFERENCE**
518 px sliding window, overlap blending, plus TTA
- 6 SCALE ANCHOR**
Copernicus GLO-30, or ground control by power-law fit
- 7 OUTPUTS**
GeoTIFF DSM, sigma map, three.js scene with a live error map

HOW IT IS VALIDATED

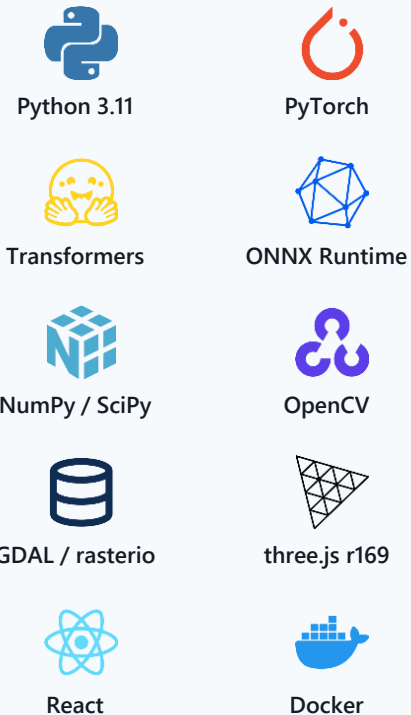
Split region-disjoint, not random — adjacent tiles share buildings.

Scored on 80 whole tiles, 11,983,760 pixels, 3,090 buildings.

Reported per urban / sparse / forested / mixed, and per height band.

Held back the test split is untouched until the very end.

TECH STACK



WHAT SHIPS

An **elevation module** emitting a GeoTIFF DSM and a sigma map, and a **viewer** that opens it and checks a height against reference data. No GPU, no network.

DECISIONS, AND WHY

- The head was chosen by measurement**
Regression, binned and head-tail-cut heads were all trained and scored. All three land at a 0.43–0.49 slope, so the simplest ships.
- Uncertainty is calibrated, not decorative**
ECE 0.077, and a sigma-vs-error rank correlation of +0.829 on held-out tiles.
- Export is checked, not assumed**
ONNX agrees with PyTorch to 0.0010 cm over 804,972 inputs. int8 costs 0.161 m for a 63% smaller file.
- Resolution is handled explicitly**
Auto-zoom reads GSD from metadata and resamples, recovering all but 3.6% of the coarse-input penalty.

**TECHNICALLY PROVEN**

Built and measured, not proposed.

**OPERATIONALLY SIMPLE**

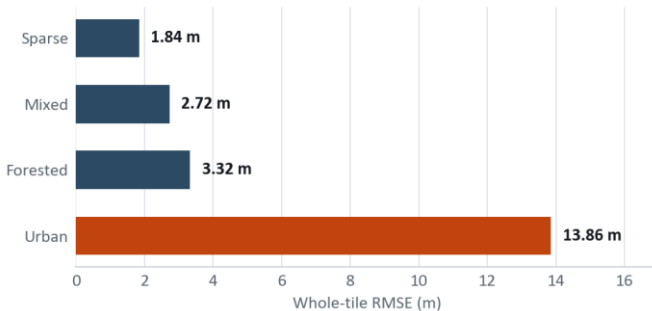
One image in. No GPU, no network.

**ECONOMICALLY VIABLE**

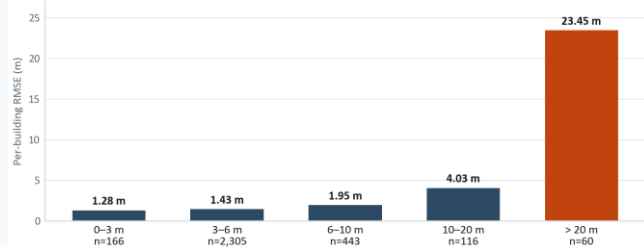
Every dependency free and open.

**SOCIALLY USEFUL**

Planning, disaster and forestry use height.

ACCURACY ACROSS THE FOUR LANDSCAPES ISRO NAMES

Three of four classes sit under 3.4 m, and correlation never drops below +0.58. The fourth is the honest problem, and it is one specific thing.

URBAN 13.86 M, DECOMPOSED

79% of squared error comes from 1.9% of buildings. Under 10 m — 94% of the stock — we are at 1.4 m. Urban RMSE is 60 tall buildings dominating a squared metric.

THE TWO REAL RISKS, AND WHAT ANSWERS THEM**No Indian ground truth exists to validate against**

Our only Indian check is Sikkim against Google Open Buildings: bias -6.94 m on 168 footprints. That is model-vs-model agreement, not accuracy, and we report it as such. No public Indian scene carries LiDAR truth.

**Tall buildings compress toward the mean**

A data problem, not an architecture one: training tops out at 82.8 m while validation reaches 155.3 m. Answered by a power-law control-point fit that cuts tall-tile RMSE 24.9%, and structurally by adding tall-building LiDAR.

WHAT REDUCES THE INDIAN-DOMAIN RISK BEFORE WE HAVE THE DATA

Resolution auto-zoom normalises GSD; measured to recover all but 3.6%.

Look angle off-nadir spans 4.8 to 28.9 degrees in DFC, so degradation is measurable on data we hold.

Terrain leakage ruled out — on 31-degree slopes, height-vs-slope correlation is -0.044.

Licensing every dependency is free and open, so nothing blocks deployment on ISRO imagery.

WHO IT SERVES



ISRO AND SAC

Height from single-view Cartosat archives, no new acquisition.



URBAN PLANNING

Building heights for FSI checks and shadow studies under AMRUT.

DEPTH WIZARD



DISASTER RESPONSE

Post-event surface models where a stereo pair cannot be waited for.



FORESTRY

Canopy height proxies for biomass and encroachment monitoring.



TELECOM ROLLOUT

Line-of-sight and tower siting without a costly LiDAR survey.

KEY BENEFITS



One image replaces a stereo pair — no revisit wait.



No GPU procurement: 36.8 MB runs on a field laptop.



Decades of existing archive gain a new product.



Municipal bodies get heights with no survey budget.



Disaster teams get a surface model in minutes.



Free and open — adoptable by any state agency.

1 image

input needed, not a stereo pair

0 GPUs

509 ms per tile on a laptop CPU

36.8 MB

the whole model, quantised, offline

4 terrains

scored separately, never averaged

RESEARCH THAT CHANGED WHAT WE BUILT

Depth Anything V2 Yang et al., NeurIPS 2024. The backbone we fine-tune.

HTC-DC Net Chen et al., TGRS 2023. Head-tail cut, distribution constraints.

Depth Any Canopy Ouaknine et al., 2024. The recipe for aerial height.

Beta-NLL Seitzer et al., ICLR 2022. Keeps the mean head learning.

IM2HEIGHT, TSE-Net Prior single-view height — our baseline for the field.

DATA AND REFERENCE SURFACES

DFC2019 Track 1 US3D at 0.3 m GSD with airborne LiDAR truth.

Copernicus GLO-30 Free 30 m global DEM; the absolute metric anchor.

GlobalBuildingAtlas Published 5.9 m RMSE over Asia — the external bar.

ISRO Bhoonidhi Registered. CartoDEM 30 m and LISS-4 are the free tiers.

Google Open Buildings A cross-check over India, never used as truth.

TRIED, MEASURED, REJECTED

Ordinal / binned head All heads land at a 0.43–0.49 slope. No gain.

Shadow photogrammetry Oracle shadows reach r 0.503; the net reaches 0.787.

LDS tail reweighting Pixels above 20 m are 7.5% but carry 78.8% of error.

Two-model ensemble Error correlation 0.925 — averaging buys nothing.

Global de-compression A rescale moves error, it does not remove it.

PROBES WE RAN, AND WHAT EACH CLOSED

01 Tall buildings More tall examples? No — training tops out at 82.8 m.

03 Guided filtering Squaring off roofs made every metric worse.

04 Where detail went Detail follows the ground area one token covers.

05 Eval resolution Cartosat-class input costs 3.6% after auto-zoom.

06 Height compression Measured: ours = $0.473 \times \text{truth} + 2.66$ m.

HOW WE COMPARE

PARAMETER	STEREO OR LIDAR SURVEY	PUBLISHED SINGLE-VIEW	DEPTHWIZARD
One archived image	✗	✓	✓
Absolute metric height	✓	—	✓
Per-pixel uncertainty	✗	✗	✓
Scored per terrain	—	✗	✓
Runs with no GPU	—	✗	✓
3D viewer shipped	✓	✗	✓
Failure modes published	—	—	✓

✓ yes — partial ✗ no

TWO ROWS THE ALTERNATIVES WIN

A stereo survey gives absolute height directly and ships mature viewers. The table is only useful to a panel if the rows we lose are still in it.

THE BAR WE ARE MEASURED AGAINST

3.62 m per-building RMSE against airborne LiDAR, versus 5.9 m published for Asia. Scored on 3,090 buildings across 80 held-out tiles.